INCH-POUND

MIL-H-53027A(ME)
AMENDMENT 4
18 December 1990
SUPERSEDING
AMENDMENT 3
19 May 1988

MILITARY SPECIFICATION

HOSE ASSEMBLY, RUBBER: LIGHTWEIGHT COLLAPSIBLE,

6-INCH: FOR DRINKING (POTABLE) WATER

This amendment forms a part of MIL-H-53027A(ME), dated 29 May 1985, and is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, line 2, delete "swivel".

Add the following new paragraph under 1.1:

* "1.2 Classification. The length of the hose assembly shall be specified ±1 percent, including the two end couplings."

PAGE 2

* "2.1.1, under SPECIFICATIONS, MILITARY, delete the following documents in their entirety.

"MIL-G-20241	-	Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	-	Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	-	Sealing and Coating Compound, Corrosion Inhibitive."

Under STANDARDS, MILITARY, delete "MIL-STD-130" in its entirety.

2.1.2, under DRAWING, ME, delete "D13225E9182" and substitute "D13227E9787". The title remains the same.

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* 2.1.2, add:

"NATIONAL SANITATION FOUNDATION (NSF)

NSF Standard 14 - Official Plastics Listing

(Application for copies should be addressed to National Sanitation Foundation, 3475 Plymouth Rd., P.O. Box 1468, Ann Arbor, MI 48106.)"

PAGE 3

2.2, under AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM), delete "ASTM B 247" in its entirety and add:

"ASTM B 26 - Aluminum Alloy Sand Castings.
ASTM B 241 - Aluminum and Aluminum Alloy Seamless Pipe and Seamless Extruded Tube."

PAGE 4

3.1, line 1, delete "500-foot".

Line 2, after "section of hose." add the following sentence: "The length of the hose assembly shall be as specified (see 6.2). A tolerance in length of 1 percent will be permitted."

Line 3, delete "This assembly shall be packed in a flaking box." and add "Two of these assemblies, connected together with a grooved coupling (see 3.5.3), shall be packed in a flaking box."

PAGE 5

* 3.4.1, 3.4.1.1, 3.4.1.2 delete in their entirety and substitute the following new paragraphs:

"3.4.1 Material deterioration prevention and control. The hose assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the hose assembly may be exposed.

3.4.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative."

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* 3.4.2.1, line 3, after "Code of Federal Regulations" add: "or the liner material and all surfaces that contact drinking water shall be tested and listed as acceptable for contact with drinking water by the National Sanitation Foundation (NSF) under Standard 14."

3.5, line 1, delete "500-foot". Delete second sentence in its entirety and add: "The length of each section, including the two end couplings, shall be as specified (see 6.2)."

3.5.1, line 5, delete "33303." and substitute "30277." Delete from "The hose shall be marked . . ." through to the end of the paragraph.

PAGES 5 AND 6

3.5.1.1, delete "Unless otherwise specified . . . shall be 500 feet ±5 feet." Add the following new sentence: "The length of each hose section, including the two end couplings, shall be as specified (see 6.2)."

PAGE 6

3.5.2, lines 3 and 4, delete sentence, "The aluminum . . . temper T6."

Add at end of paragraph: "The aluminum shall conform to ASTM B 241, alloy A96061, temper T6, or ASTM B 26, alloy UNS A 03560, temper T6. The clamps shall be non-vivid and non-reflective."

3.6.1.3, line 2, delete "craking" and substitute "cracking".

PAGE 7

3.6.7, line 1, delete "(500-foot)".

Lines 3 and 4, delete "1/2 turn per hose section", and substitute: "36 degrees per 100 feet of length".

3.7, delete the entire paragraph and substitute:

"3.7 Identification marking. Each hose assembly shall have the following information:

Date of manufacturer (quarter and year).
Contract or order number.
Manufacturer's name and trademark.
The words "DRINKING WATER ONLY."

Hoses shall be branded at intervals not greater than 10 feet. The letters shall be not less than 0.18-inch high. Label color shall contrast with hose cover color. Each hose assembly shall be marked with a longitudinal blue stripe not less than 0.125-inch wide and extending the full length of the hose. The label may substitute for the longitudinal stripe, provided it is continuous for the full length of the hose and is blue in color. The label may be within the longitudinal stripe provided the width of the stripe is not less than 0.20-inch greater than the height of the label letters, and the

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label is in contrast to the color of the stripe. The label and stripe shall remain clear, distinct and legible for the life of the hose and shall not be adversely affected by handling, bending, water, oil, or environmental effects."

Add new paragraph as follows:

"3.10 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the hose assemblies may be newly fabricated from recovered materials to the maximum extent practicable, provided the hose assemblies produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the hose assemblies."

PAGE 9

4.4.1.2.3, line 2, delete "4.0 percent" and substitute "1.0 percent".

4.4.2, line 2, delete "2.5 percent" and substitute "1.0 percent".

4.5.2.2.1, line 3, delete "3.65" and substitute "3.6.6".

4.5.2.2.2, line 2, delete "500" and substitute "100".

Line 8, delete "3.6.6" and substitute "3.6.7".

4.5.2.3, delete last sentence "Nonconformance to 3.6.2, 3.6.3, or 3.6.4 shall constitute failure of these tests."; and substitute "Nonconformance to 3.6.2, 3.6.3, 3.6.4, and 3.6.5 shall constitute failure of these tests."

PAGE 10

* 4.5.1, under defects, add:

"118. Used, rebuilt, or remanufactured components pieces, or parts incorporated into the hose assembly."

4.5.2.3.1, line 5, after "70-hour test.", add: "Maintain sample filled with water at 82 °C for 30 minutes, then allow to cool to 37 °C. Maintain at 37 °C for the remaining 69.5 hours."

PAGE 14

6.2, add the following:

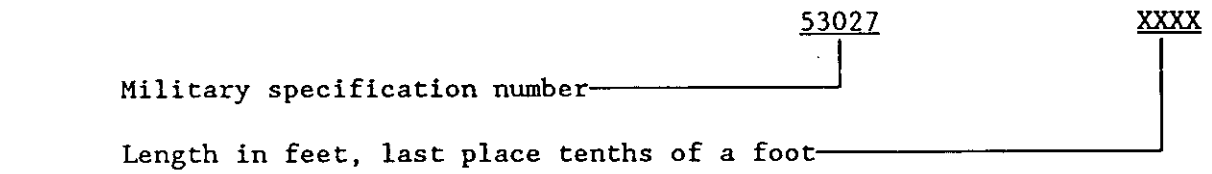
"g. Specification part number required (see 1.2.1).
h. Length of hose assembly required (see 3.1).
i. Length of each hose section required (see 3.5 and 3.5.1.1)."

6.5, delete in its entirety.

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* Add new paragraph 6.5, as follows:

"6.5 Part or identifying number. The part or identifying number for the hose assemblies as specified (see 6.2) shall be in feet with the last number expressed in tenths of a foot and shall be designated as follows:



EXAMPLE: M53027-5000, Hose Assembly, per MIL-H-53027, 500.0 feet long".

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